# Moore, IPS 6e Chapter 05

*Quizzes prepared by Dr. Patricia Humphrey, Georgia Southern University*

Suppose that you are a student worker in the Statistics Department and they agree to pay you using the Random Pay system. Each week the Chair flips a coin. If it comes up heads, your pay for the week is $80, and if it comes up tails, your pay for the week is $40. You work for the department for two weeks. Let $\bar{x}$ be the average of the pay you receive for the first and second week. The sampling distribution of $\bar{x}$ is

$\bar{x}$: $40 $60 $80

- a. given by the following probability distribution.
  Probability: 0.25 0.50 0.25
- b. approximately normal with mean $60 and standard deviation 14.14.
- c. exactly normal with mean $60 and standard deviation 14.14.

---

The scores of individual students on the American College Testing (ACT) Program composite college entrance examination have a normal distribution with mean that varies slightly from year to year and standard deviation 6.0. You plan to take an SRS of size *n* of the students who took the ACT exam this year and compute the mean score $\bar{x}$ of the students in your sample. You will use this to estimate the mean score of all students this year. In order for the standard deviation of $\bar{x}$ to be no more than 0.1, how large should *n* be?

- a. At least 60.
- b. At least 3600.
- c. This cannot be determined because we do not know the true mean of the population.

---

When a particular penny is held on its edge and spun, the probability that heads are face up when the coin comes to rest is 4/9. If the coin is spun 4 times, the probability that the coin will come up heads exactly twice is (assume trials are independent)

- a. 16/81.
- b. 0.366.
- c. 0.061.

---

You decide to test a friend for ESP using a standard deck of 52 playing cards. Such a deck contains 13 spades, 13 hearts, 13 diamonds, and 13 clubs. You shuffle the deck, select a card at random, and ask your friend to tell you whether the card is a spade, heart, diamond, or club. After the guess you return the card to the deck, shuffle the cards, and repeat the above. You do this a total of 100 times. Let *X* be the number of correct guesses by your friend in the 100 trials. The standard deviation of *X* is

- a. 0.433.

- b. 4.33.
- c. 18.75.

---

You decide to test a friend for ESP using a standard deck of 52 playing cards. Such a deck contains 13 spades, 13 hearts, 13 diamonds, and 13 clubs. You shuffle the deck, select a card at random, and ask your friend to tell you whether the card is a spade, heart, diamond, or club. After the guess you return the card to the deck, shuffle the cards, and repeat the above. You do this a total of 100 times. Let *X* be the number of correct guesses by your friend in the 100 trials. What is the probability that $X \geq 30$ if your friend is just guessing? Use the normal approximation with the continuity correction.

- a. 0.1492
- b. 0.30
- c. 0.8508

---

Which of the following might be reasonably modeled by the binomial distribution?

- a. The number of customers that enter a store in a one-hour period, assuming customers enter independently.
- b. The number of questions you get correct on a 100-question multiple choice exam in which each question has only four possible answers. Assume you have studied extensively for the test.
- c. None of the above.

---

A set of ten cards consists of five red cards and five black cards. The cards are shuffled thoroughly and I am given the first four cards. I count the number of red cards *X* in these four cards. The random variable *X* has which of the following probability distributions?

- a. The binomial distribution with parameters $n = 10$ and $p = 0.5$.
- b. The binomial distribution with parameters $n = 4$ and $p = 0.5$.
- c. None of the above.

---

A high school has 1000 students. As a school project, the school conducts a mock lottery. Each student is the school is asked to select an integer between 1 and 1000, independently of the choices of the other students. Each student gives their selection, with their name, to the school principal. The school principal uses a random number generator to select an integer between 1 and 1000 and any student that picked this number is declared a winner. What is the probability that no student wins?

- a. 0. There are 1000 students and 1000 numbers, so some student must win.
- b. $(0.999)^{1000}$ .
- c. 0.1587.

---

There are twenty multiple-choice questions on an exam, each having responses a, b, c, or d. Each question is worth 5 points and only one response per question is correct. Suppose a student *guesses* the answer to each question, and her guesses from question to question are independent. If the student needs at least 40 points to pass the test, the probability the student passes is closest to

- a. 0.0609.
- b. 0.1019.
- c. 0.9590.

---

Suppose that scores on the Math SAT exam follow a normal distribution with mean 500 and standard deviation 100. Two students that have taken the exam are selected at random. What is the probability that the sum of their scores exceeds 1200?

- a. 0.1587
- b. $(0.1587)^2$
- c. 0.0793

---

There are twenty multiple-choice questions on an exam, each having responses a, b, c, or d. Each question is worth 5 points and only one response per question is correct. Suppose a student *guesses* the answer to each question, and her guesses from question to question are independent. The student's mean *score on the exam* should be

- a. 5.
- b. 25.
- c. 50.

---

The scores of individual students on the American College Testing (ACT) Program College Entrance Exam have a normal distribution with mean 18.6 and standard deviation 6.0. At Westside High, 400 students take the test. Assume their scores are all independent. If the scores at this school have the same distribution as nationally, the probability that the average of the scores of all 400 students exceeds 19.0 is

- a. the same as the probability that a single student has a score exceeding 19.0.
- b. larger than the probability that a single student has a score exceeding 19.0.
- c. smaller than the probability that a single student has a score exceeding 19.0.

---

As part of a promotion for a new type of cracker, free samples are offered to shoppers in a local supermarket. The probability that a shopper will buy a packet of crackers after tasting the free sample is 0.200. Different shoppers can be regarded as independent trials. Let $\hat{p}$ be the proportion of the next *n* shoppers that buy a packet of the crackers after tasting a free sample. How large should *n* be so that the standard deviation of $\hat{p}$ is no more than 0.01?

- a. 4.

- b. 16.
- c. 1600

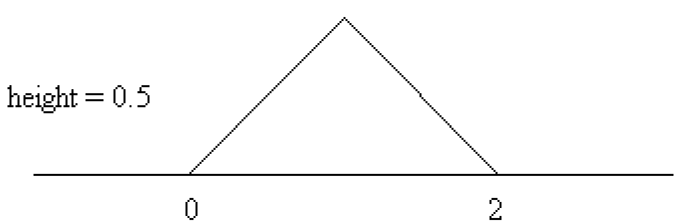
---

Suppose that the random variable *X* has a distribution with a density curve that looks like the following.

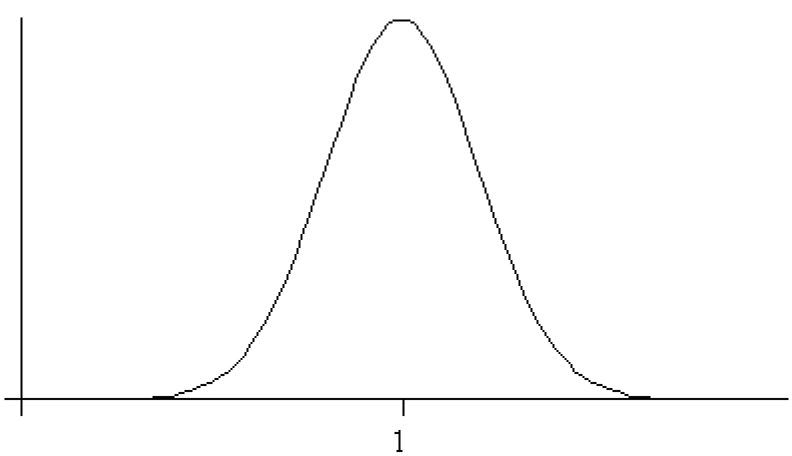

The sampling distribution of the mean of a random sample of 100 observations from this distribution will have a density curve that looks most like which of the following?

- a.


- b.




- c.



As part of a promotion for a new type of cracker, free trial samples are offered to shoppers in a local supermarket. The probability that a shopper will buy a packet of crackers after tasting the free sample is 0.200. Different shoppers can be regarded as independent trials. If $\hat{p}$ is the proportion of the next 100 shoppers that buy a packet of the crackers after tasting a free sample, then $\hat{p}$ has approximately a

- a. *N* (0.2, 0.0016) distribution.
- b. *N* (0.2, 0.04) distribution.
- c. *N* (0.2, 4) distribution.

---

As part of a promotion for a new type of cracker, free samples are offered to shoppers in a local supermarket. The probability that a shopper will buy a packet of crackers after tasting the free sample is 0.200. Different shoppers can be regarded as independent trials. If $\hat{p}$ is the proportion of the next 100 shoppers that buy a packet of the crackers after tasting a free sample, then the probability that fewer than 30% buy a packet after tasting a free sample is approximately (don't use the continuity correction)

- a. 0.3000.
- b. 0.9938.
- c. None of the above.

---

Suppose that you are a student worker in the statistics department and agree to be paid by the Random Pay system. Each week the Chair flips a coin. If the coin comes up heads, your pay for the week is $80; if it comes up tails, your pay for the week is $40. You work for the department for 100 weeks (at which point you have learned enough probability to know the system is not to your advantage). The probability that $\bar{x}$, your average earnings in the first *two* weeks, is greater than $65 is

- a. 0.2500.
- b. 0.3333.
- c. 0.5000.

---

The scores of individual students on the American College Testing (ACT) Program composite college entrance examination have a normal distribution with mean 18.6 and standard deviation 6.0. At Northside High, 36 seniors take the test. If the scores at this school have the same distribution as national scores, the sampling distribution of the average (sample mean) score for the 36 students is

- a. approximately normal, but the approximation is poor.
- b. approximately normal, and the approximation is good.
- c. exactly normal.

---

Suppose that you are a student worker in the statistics department and agree to be paid by the Random Pay system. Each week the Chair flips a coin. If the coin comes up heads, your pay for the week is $80; if it comes up tails, your pay for the week is $40. You work for the department for 100 weeks (at which point you have learned enough probability to know the system is not to your advantage). Suppose $\bar{x}$ is your average pay for the 100 weeks. Then $\bar{x}$ has approximately a

- a. *N* (60, 10.95) distribution.
- b. *N* (60, 2) distribution.
- c. *N* (120, 20) distribution.

---

The SAT scores of entering freshmen at University X have a *N* (1200, 90) distribution and the SAT scores of entering freshmen at University Y have a *N* (1215,110) distribution. A random sample of 100 freshmen is sampled from each University, with $\bar{x}$ the sample mean of the 100 scores from University X and $\bar{y}$ the sample mean of the 100 scores from University Y. The probability that $\bar{x}$ is greater than μY, the population mean for University Y, is

- a. 0.0475.
- b. 0.0869.
- c. 0.4325.

---

Incomes in a certain town are strongly right skewed with mean $36000 and standard deviation $7000. A random sample of 10 households is taken. What is the probability the average of the sample is more than $38000?

- a. 0.3875
- b. 0.1831.
- c. Cannot say.

---

Incomes in a certain town are strongly right skewed with mean $36000 and standard deviation $7000. A random sample of 75 households is taken. What is the standard deviation of the sample mean?

- a. $808.29
- b. $93.33
- c. $7000.

---

Incomes in a certain town are strongly right skewed with mean $36000 and standard deviation $7000. A random sample of 75 households is taken. What is the probability the sample mean is greater than $37000?

- a. 0.4432
- b. 0.1075
- c. 0

Which of the following would have a Binomial Distribution?

a. A couple will have children until they have three girls or five children. X is the number of children in the family.

b. Max the magician (who practices flipping coins) will call flips of a coin in the air. X is the number of correct calls in 10 flips.

c. None of the others is Binomial.

In a pre-election poll, 400 of the 500 probable voters polled favored the incumbent. In this poll, the sample proportion, $\hat{p}$, of those favoring the challenger is

a. 0.80

b. 0.20

c. 0.50

The binomial coefficient, written $\binom{n}{k}$, gives what information?

a. The probability of obtaining *k* successes in *n* trials.

b. The number of ways in which one can obtain k successes in n trials

c. The chance of observing a success.

From previous polls, it is believed that 66% of likely voters prefer the incumbent. A new poll of 500 likely voters will be conducted. In the new poll, if the proportion favoring the incumbent has not changed, what is the mean and standard deviation of the number preferring the incumbent?

a. $\mu = 330$, $\sigma = 10.59$

b. $\mu = 0.66$, $\sigma = 0.021$

c. $\mu = 330$, $\sigma = 18.17$

From previous polls, it is believed that 66% of likely voters prefer the incumbent. A new poll of 500 likely voters will be conducted. In the new poll, if the proportion favoring the incumbent has not changed, what is the probability that more than 68% will favor the incumbent?

a. 32%.

b. 82.7%

c. 17.1%

It has been asserted that 15% of all people are left-handed. In a random group of 10 people, what is the chance that not all of them are right-handed?

a. 85%

b. 80.3%

c. 100%

---

Sodas in a can are supposed to contain an average 12 oz. This particular brand has a standard deviation of 0.1 oz, with an average of 12.1 oz. If the cans contents follow a normal distribution, what is the probability that the mean contents of a six-pack are less than 12 oz?

- a. 0.007
- b. 0.1587
- c. 0.9928

---

We want to take a sample of 100 items out of a large batch for quality control purposes. Based on past history, the proportion of defective items is 4%. Can we use the normal approximation to the binomial distribution to find the probability of finding more than 5 defective items in the sample of 100?

- a. Yes, because *n* is large.
- b. No
- c. We do not have enough information.

---

If I select a random sample of 20 items from a large batch in which there are 5% defective items, what is the chance I find at most two defectives in the sample?

- a. 0.1887
- b. 0.9246
- c. 0.0754

---

A national poll of 600 men announced that the proportion in the survey who claimed to help their wives at home was 85%. If we took a larger poll of 1200 men, about how many would we expect to say they help their wives at home, based on the first survey?

- a. 510.
- b. 600
- c. 1020

---

A national poll of 600 men announced that the proportion in the survey who claimed to help their wives at home was 85%. If we took a larger poll of 1200 men, what will be the standard deviation of the number of men who help at home, based on the first survey?

- a. 12.37
- b. 17.32
- c. 153

---

A machine fills cans of soda which are labeled "12 ounces" according to a normal distribution with mean 12.1 ounces and standard deviation 0.1 ounces. If I buy a 12-pack of the soda, what is the chance the average contents of the cans will be less than advertised?

- a. 0.1587

- b. 0.0003
- c. 0.50

---

Submit